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Defect Detection Using High-Resolution Dual Caliper across Multiple Pipes in Single Descent

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Abstract

Traditionally, multifinger caliper tools are designed for specific pipe sizes, necessitating the use of multiple tools with different outer diameters to accommodate varying pipe dimensions. This design limitation prevents a single tool from being deployed across multiple pipe sizes in a single descent, thereby requiring multiple runs. Additionally, traditional multifinger calipers often suffer from limited circumferential coverage, which often results in failure to detect smaller defects such as pinholes and minor pitting.

This paper presents a novel high-resolution dual caliper designed to address challenges in detecting defects within downhole tubulars, completion components, and casings. The innovative tool features an 80-finger arrangement within a compact 2 1/8-inch outer diameter, enabling it to cover a broad range of pipe sizes, from 2 5/8-inch tubing to 10-inch casing, without compromising radial measurement accuracy. Eccentricity effects are minimized through the equal azimuthal spacing of the fingers, supported by a specialized data processing algorithm. For highly deviated wells, motorized centralizers ensure precise tool centering, further enhancing measurement reliability.

As part of this study, multiple wells were successfully logged using the novel high-resolution dual caliper technology, capable of inspecting various pipe sizes in a single descent. The dual caliper's enhanced circumferential coverage and higher inner radius resolution significantly improved the detection of minor defects such as localized pitting and pinholes, outperforming traditional multifinger calipers in accuracy and resolution. The dual caliper's advanced data processing incorporates specialized merge algorithms, enabling seamless visualization of data from all 80 fingers. Advanced data processing facilitates rapid interpretation of critical zones, supporting informed decision-making for subsequent operations. The processing algorithms also provide quick-look diagnostics, including maximum penetration depth, metal loss per joint, and minimum inner diameter, ensuring efficient evaluation of pipe integrity. The field performance of this novel technology demonstrates a significant advancement in downhole evaluation technology, delivering reliable and actionable insights. By delivering precise and reliable data, this advanced technology equips exploration and production (E&P) companies with critical insights for designing effective intervention strategies.

The multi-well study has proven invaluable in enabling E&P companies to better assess the condition of internal tubing and casing, facilitating timely and informed decisions regarding necessary remedial actions. The novel high-resolution dual caliper tool demonstrated its efficiency by evaluating multiple pipe sizes in a single descent, significantly reducing operational time and costs. The dual caliper's ability to provide the most accurate information enhances well integrity management, ensuring optimal performance and long-term asset reliability.

Introduction

Pipe condition assessment remains a critical component in the lifecycle management of oil and gas wells. Accurate and timely identification of defects such as corrosion, pitting, and pinholes is essential to ensure long-term zonal isolation and mechanical stability. Traditionally, multifinger caliper (MFC) tools have been the industry standard for internal casing and tubing inspection. However, their limitation in terms of outer diameter and finger count means they are typically optimized for single pipe sizes, requiring multiple logging runs to evaluate different tubulars. This limitation not only increases rig time and operational cost but also delays the diagnosis and remediation of critical issues.

In response to these challenges, a new high-resolution dual caliper has been developed with the capability to assess a wide range of tubular sizes in a single descent, while enhancing circumferential resolution and defect detection accuracy. This paper presents the field application, performance analysis, and advantages of this novel tool, including comparisons with conventional multifinger caliper logging techniques.

Background and Challenges in Traditional Caliper Logging

Multifinger calipers typically consist of 24 to 60 radial arms (fingers) that contact the inner wall of the pipe to measure its internal geometry. While effective for gross deformation or wall loss detection, these tools suffer from two main drawbacks:

- Limited circumferential resolution due to fewer contact points and wide azimuthal spacing, often resulting in missed small-scale anomalies.
- Restricted operational range, necessitating separate tools for different pipe sizes, increasing the complexity and cost of well evaluations.

These limitations become especially significant in completions involving multiple pipe sizes, such as wells with nested tubing and casing, or where downhole hardware varies in diameter.

Technology Description

Tool Architecture

The high-resolution dual-caliper tool utilizes an array of hard-faced measurement fingers engineered to deliver precise internal radius measurements, even under challenging downhole environments. The tool features 80 equally spaced fingers housed within a compact 2 1/8-inch outer diameter body. This configuration allows the tool to operate effectively in pipe sizes ranging from 2 5/8-inch tubing to 10-inch casing. The dense azimuthal finger spacing increases circumferential coverage, improving the detection of localized anomalies that could otherwise be missed.

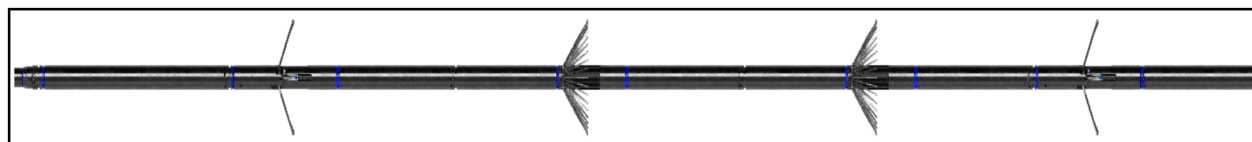


Figure 1—The high-resolution dual caliper is the industry's first and smallest OD tool featuring 80 fingers that provide inner pipe coverage to increase the chances of detecting small defects

The tool features a modular architecture composed of dual 40-finger caliper modules, enabling operational flexibility. This design allows for the removal of one module to accommodate surface rig-up height restrictions or specific downhole constraints, without compromising performance.

Centering and Eccentricity Mitigation

Eccentricity effects are minimized by the equal azimuthal spacing of the fingers and a special processing algorithm. To ensure consistent measurements, especially in deviated or horizontal wells, motorized centralizers are integrated to maintain tool centering within the wellbore. This minimizes eccentricity effects, a common source of error in radial measurements.

Data Acquisition and Processing

Advanced algorithms merge data from all 80 fingers to generate high-resolution cross-sectional images. The processing suite includes automatic computation of key metrics such as:

- Maximum metal loss and its depth
- Metal loss per joint
- Burst pressure per joint
- Minimum internal diameter

The tool also supports real-time quick-look diagnostics to assist E&P companies in identifying zones requiring immediate attention.

Field Deployment

Operational Overview

The tool was deployed across multiple wells with diverse completion architectures. In each case, the tool successfully logged through multiple casing and tubing sizes in a single descent. This reduced rig time, eliminated the need for multiple tools or tool change-outs, and simplified the planning and execution of the pipe evaluation.

Detection of Defects

Field data revealed numerous small-scale defects including pitting, pinholes, and corrosion. The tool provided high-fidelity measurements confirming the presence and geometry of these anomalies with improved spatial resolution.

Comparative Evaluation

A direct comparison was conducted between the high-resolution dual caliper and the benchmark 40-finger caliper tool in two different wells. While both tools captured the general presence of the defect, the benchmark tool underrepresented the geometry due to its limited circumferential resolution and wider azimuthal spacing. In contrast, high-resolution dual caliper provided a much sharper delineation of the feature, capturing the shape with higher accuracy. Notably, the dual-caliper design allowed for enhanced

diagnostic confidence without requiring a repeat pass. Since each of the two 40-finger caliper sections is azimuthally offset, their measurements served to validate each other, eliminating the need for repeat pass. This not only reduced rig time but also improved interpretation consistency.

Example 1: Well-A

High-resolution dual caliper tool was run in well A to provide the condition of the 7-inch liner and 9 5/8-inch casing. The modular tool consists of two sections of 40 finger calipers placed a fixed distance apart from each other and the fingers of the first caliper are shifted by 4.5 degree compared to fingers of second caliper. Also, legacy 40-finger caliper tool was run for comparison purposes.

Log Quality

In general, no quality issues were observed in the data. These are the highlights:

- i. All fingers are in contact with the inner pipe wall.
- ii. Tool eccentricity is good.
- iii. Average diameter approximated to theoretical nominal diameter in good pipe section.
- iv. No sticking or reconstructed fingers issue observed.
- v. Repeat pass showed good repeatability with main pass.

Interpretation Summary

High-resolution dual caliper data showed most of the 7-inch liner joints and 9 5/8-inch casing joints have light metal loss. The statistics for maximum recorded internal radius per joint, depth of maximum recorded internal radius per joint, maximum metal loss per joint, minimum internal diameter per joint, depth of minimum internal diameter per joint, average internal diameter per joint, maximum internal diameter per joint and maximum scale per joint were presented.

The following example shows high resolution dual caliper data across 9 5/8-inch casing and 7-inch liner. Figures 2, 4, 6 are presented with the following tracks (left to right):

1. Depth track
2. Tension (TENS), Tool Eccentricity (ECCE), Cable Speed (CS), Casing Collar Locator (CCLD) and Gamma Ray (GR)
3. Nominal Internal Radius (CIRC_1), Nominal External Radius (CORC_1), Internal Radius Average, Minimum and Maximum for top 40 fingers caliper (IRAV_1, IRMN_1, IRMX_1)
4. Individual finger traces for top 40 fingers caliper (CRAD_1)
5. Internal radius differential map for top 40 fingers caliper (CRAM_1)
6. Nominal Internal Radius (CIRC_2), Nominal External Radius (CORC_2), Internal Radius Average, Minimum and Maximum for bottom 40 fingers caliper (IRAV_2, IRMN_2, IRMX_2)
7. Individual finger traces for bottom 40 fingers caliper (CRAD_2)
8. Internal radius differential map for bottom 40 fingers caliper (CRAM_2)
9. Depth track
10. Merged Internal Radius Average, Minimum and Maximum for 80 fingers (IRAV_M, IRMN_M, IRMX_M)
11. Merged Internal radius differential map for 80 fingers (CRAM_Merged)

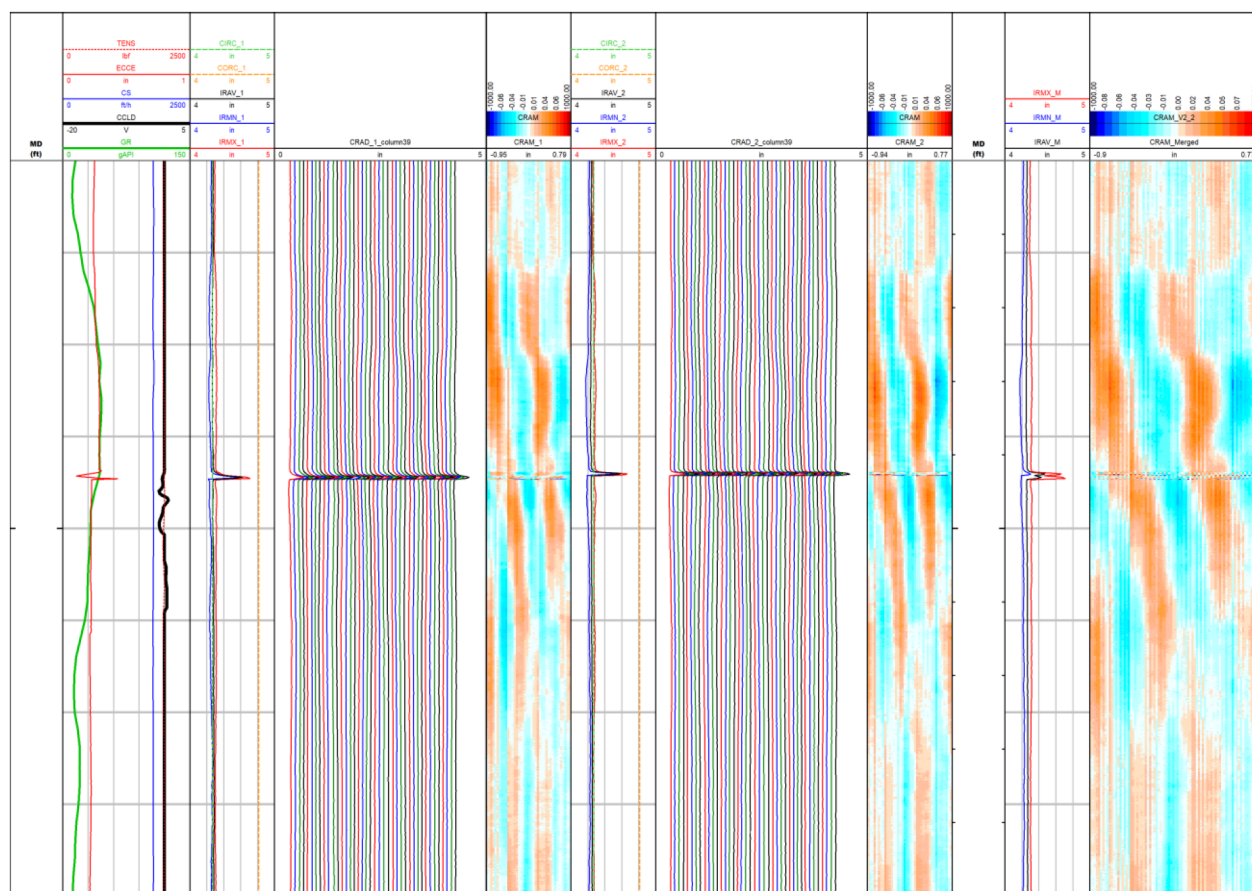


Figure 2—The high-resolution dual caliper detected slightly ovalized pipe on both calipers across 9 5/8-inch casing.

The following example also shows 40 finger caliper data across 9 5/8-inch casing and 7-inch liner. Figures 3 and 5 are presented with the following tracks (left to right):

1. Depth track, Speed (LSPD), Tool Eccentricity (CENTOFF), Gamma Ray (GR), Casing Collar Locator (CCL) and deviation (MITDEV)
2. Tension (Lten)
3. Internal Radius Average, Minimum and Maximum for top 40 fingers caliper (AVERAD, MINRAD, MAXRAD)
4. Individual finger traces for top 40 fingers caliper (FING01, FING02....FING40)
5. Internal radius differential map for top 40 fingers caliper (FINGVD)

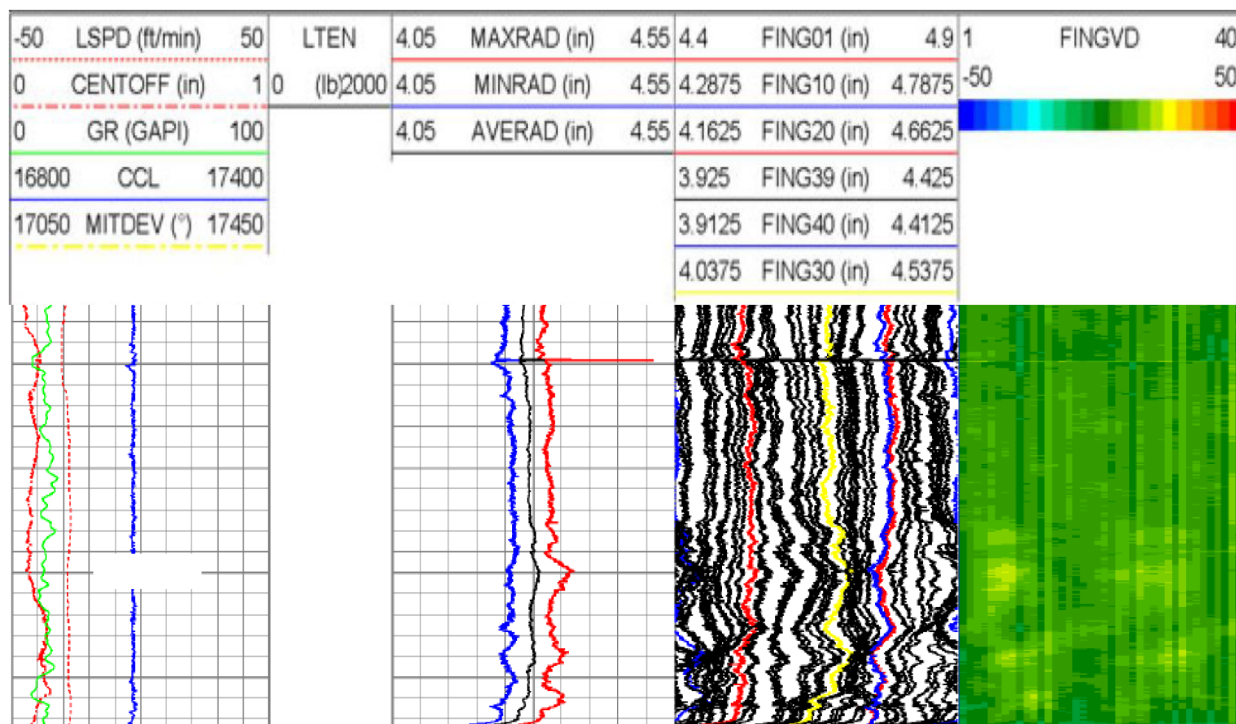


Figure 3—The benchmark caliper detected slightly ovalized pipe across 9 5/8-inch casing.

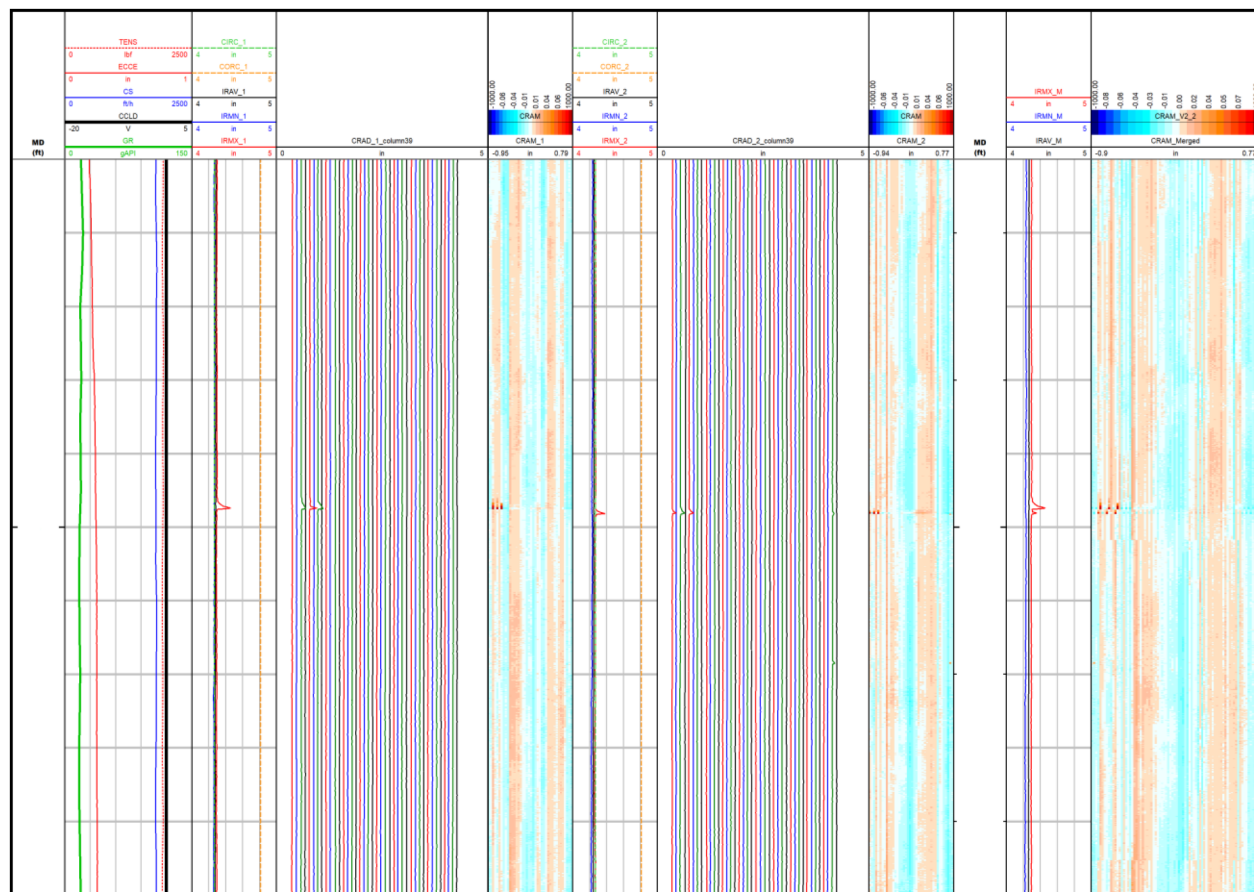


Figure 4—The high-resolution dual caliper detected localized anomaly on 3 fingers of both calipers across 9 5/8-inch casing. This is the point of highest internal radius measurement.

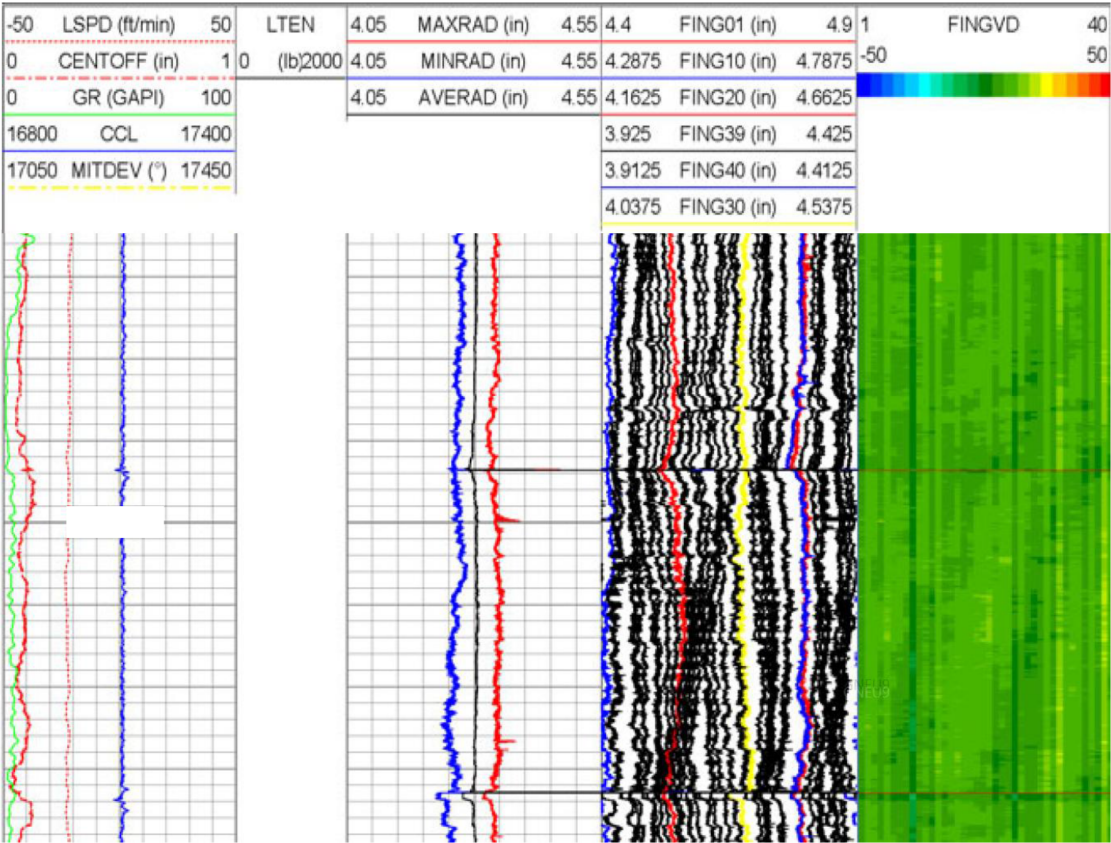


Figure 5—The benchmark caliper did not detect a localized anomaly clearly on 3 fingers across 9 5/8-inch casing.

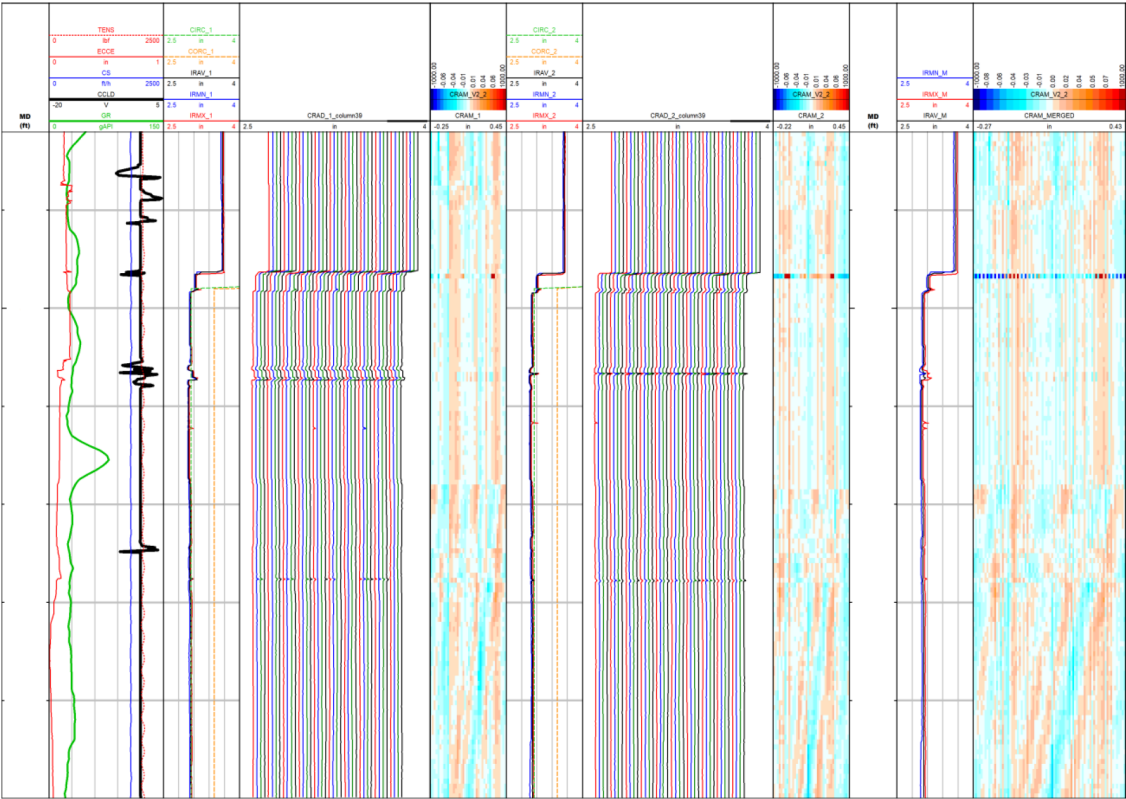


Figure 6—The high-resolution dual caliper log across 7-inch liner hanger.

In one of the evaluated intervals, both the high-resolution dual caliper and the benchmark 40-finger tool were able to detect casing ovalization. However, due to its higher circumferential resolution and denser finger coverage, high-resolution dual caliper provided a much clearer and more detailed representation of the anomaly. The ovalized geometry was more distinctly profiled in high-resolution dual caliper data, enabling better characterization of the pipe's structural condition. [Figure 2](#) shows the ovalization as detected by the high-resolution dual caliper tool, while [figure 3](#) presents the same interval as captured by the benchmark 40-finger caliper. The comparison highlights the enhanced clarity and diagnostic confidence achieved through the higher spatial resolution of the dual caliper.

In a nominally circular pipe, the internal radius readings from the fingers would be uniform across all azimuthal angles at a given depth. However, in the presence of ovalization, the cross-section of the pipe deviates from its ideal circular shape, typically deforming into an ellipse due to mechanical loading, buckling, or installation-induced stress. This deformation is directly observable in the high-resolution dual caliper data as systematic variation in internal radius readings across the circumferential array. For an ovalized pipe, the caliper fingers record a minimum internal radius along the minor axis and a maximum radius along the major axis of the oval.

The high resolution and precision readings of this 80-finger caliper tool allow for reliable detection of even slight ovalities, which are critical for assessing structural issues, collapse resistance, and tool passability in downhole environments as ovalized pipe can significantly affect tool conveyance, posing risks to the successful deployment of intervention or logging tools.

In a separate interval, high-resolution dual caliper detected a distinct localized anomaly appearing on multiple adjacent fingers, indicating a small-scale defect. This anomaly was not clearly captured by the benchmark 40-finger tool due to its lower spatial resolution and wider azimuthal spacing, which limited its ability to detect features of this scale and orientation. The increased finger density and full circumferential coverage of high-resolution dual caliper allowed for a more reliable identification and detailed characterization of the anomaly. [Figure 4](#) shows the localized anomaly as recorded by high-resolution dual caliper, while [figure 5](#) presents the same depth from the benchmark tool, where the feature appears much less defined.

The high-resolution dual caliper was able to clearly detect the 7-inch liner hanger. The 80-finger high-resolution dual caliper tool effectively identified this downhole hardware based on the distinct change in internal geometry. A liner hanger is a mechanical device used to suspend a liner string inside a parent casing and is typically located at the liner top, where an abrupt reduction in internal diameter is observed. Accurate detection and depth positioning of the liner hanger are critical for correlating well construction elements, planning future interventions, and verifying completion schematics. [Figure 6](#) shows the liner hanger response as captured by the high-resolution dual caliper tool.

Example 2: Well-B

High-resolution dual caliper tool was run in well B to provide the condition and integrity of the 4 1/2in tubing and 4 1/2in liner. The modular tool consists of two sections of 40 finger calipers placed a fixed distance apart from each other and the fingers of the first caliper are shifted by 4.5 degree compared to fingers of second caliper. Also, benchmark 40-finger caliper tool was run for comparison purposes.

Log Quality

In general, no quality issues observed in data. These are the highlights:

- i. All fingers are in contact with the inner pipe wall.
- ii. Tool eccentricity is good.
- iii. Average diameter approximated to theoretical nominal diameter in good pipe section.
- iv. No sticking or reconstructed fingers issue observed.

- v. Repeat pass showed good repeatability with main pass.

Interpretation Summary

High-resolution dual caliper data showed most of the 4 1/2in tubing and 4 1/2in liner joints have light metal loss. The statistics for maximum recorded internal radius per joint, depth of maximum recorded internal radius per joint, maximum metal loss per joint, minimum internal diameter per joint, depth of minimum internal diameter per joint, average internal diameter per joint, maximum internal diameter per joint and maximum scale per joint were presented.

The following example shows high resolution dual caliper data across 4 1/2in tubing and 4 1/2in liner. [Figures 7](#) and [9](#) are presented with the following tracks (left to right):

1. Depth track
2. Tension (TENS), Tool Eccentricity (ECCE), Cable Speed (CS), Casing Collar Locator (CCLD) and Gamma Ray (GR)
3. Nominal Internal Radius (CRIC_1), Nominal External Radius (CORC_1), Internal Radius Average, Minimum and Maximum for top 40 fingers caliper (IRAV_1, IRMN_1, IRMX_1)
4. Individual finger traces for top 40 fingers caliper (CRAD_1)
5. Internal radius differential map for top 40 fingers caliper (CRAM_1)
6. Nominal Internal Radius (CRIC_2), Nominal External Radius (CORC_2), Internal Radius Average, Minimum and Maximum for bottom 40 fingers caliper (IRAV_2, IRMN_2, IRMX_2)
7. Individual finger traces for bottom 40 fingers caliper (CRAD_2)
8. Internal radius differential map for bottom 40 fingers caliper (CRAM_2)
9. Depth track
10. Merged Internal Radius Average, Minimum and Maximum for 80 fingers (CM_IRAV_M, CM_IRMN_M, CM_IRMX_M)
11. Merged Internal radius differential map for 80 fingers (CM_CRAM_M)

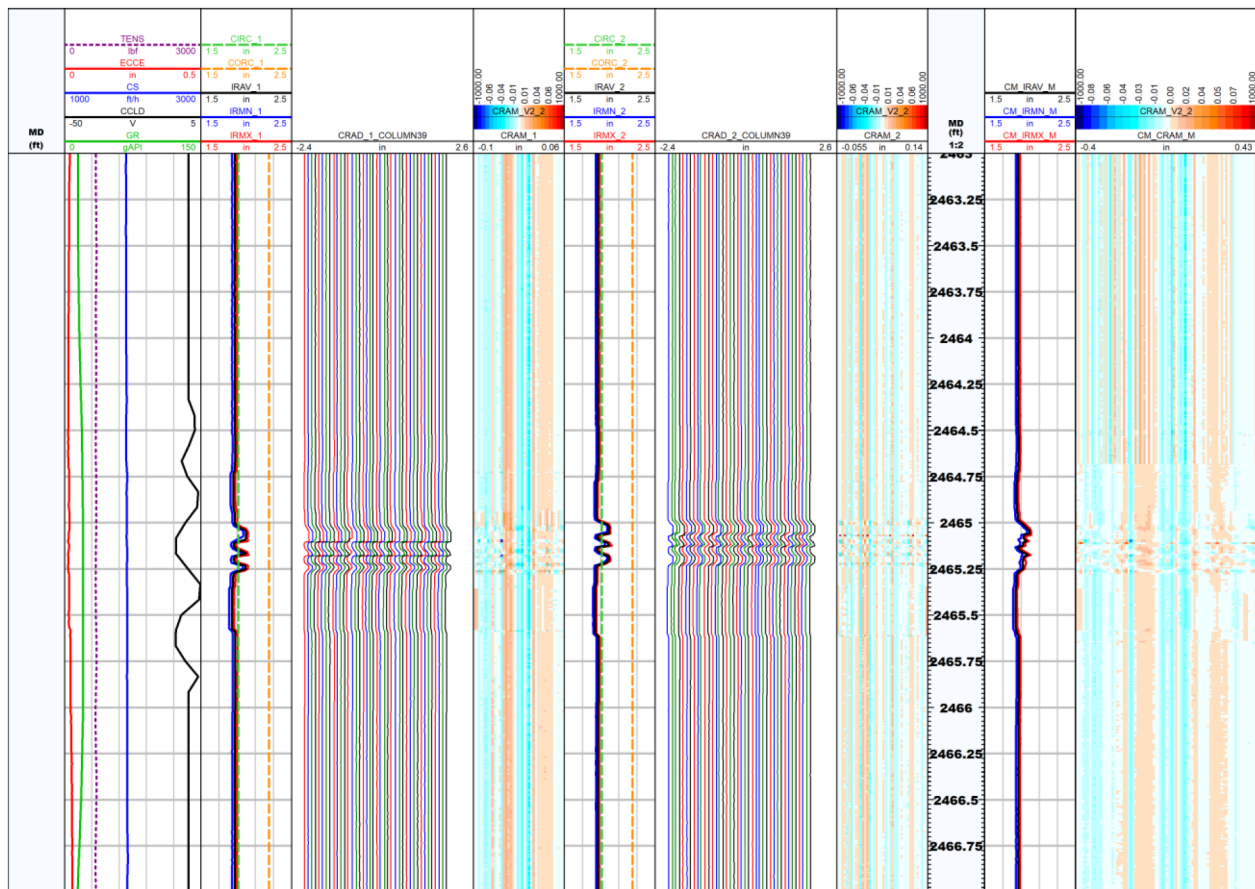


Figure 7—The high-resolution dual caliper detected R-Nipple on both calipers across 4 1/2in tubing.

The following example also shows 40 finger caliper data across 4 1/2in tubing and 4 1/2in liner. Figure 8 is presented with the following tracks (left to right):

1. Depth track,
2. Internal Radius Average, Minimum and Maximum for top 40 fingers caliper (Average Radius, Minimum Radius, Maximum Radius), Nominal Internal Radius (NOM_IR), Nominal External Radius (NOM_OR), Tool Eccentricity (Eccentricity), Calculated Metal Loss (Metal Loss)
3. Individual finger traces for top 40 fingers caliper
4. Speed (LSPD), Cable Tension (Cable Tension), Deviation (DEVI), Pressure (PTC_PRES), Temperature (PTC_WTEMP), Casing Collar Locator (PTC_CCL)
5. Borehole shape (BOREHOLE_ML)
6. Calculated Metal Loss (Percent Metal Loss)



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interventions. Figure 7 shows the R-nipple response from the high-resolution dual caliper tool, while Figure 8 presents the same feature as logged by the benchmark caliper.

In Fig. 9, high resolution dual caliper data showcases a clear crossover from 4 1/2in tubing to 7in casing, identified through a sudden and sustained change in the measured internal radius profile. The 80-finger caliper tool, designed to capture high-resolution radial geometry along the wellbore, recorded a distinct increase in internal radius across the crossover. From a diagnostic perspective, confirming the exact depth and integrity of tubing to casing crossover is essential for completion verification, well integrity assurance and planning future interventions.

Results and Discussion

The field results clearly demonstrate the tool's capability to detect both widespread and localized damage in a wide range of pipe diameters with minimal operational disruption. Key benefits observed include:

- Improved defect characterization due to high-resolution circumferential imaging
- Operational efficiency through single pass logging across varying pipe sizes
- Cost reduction by eliminating the need for multiple descents and tool swaps
- Improved decision-making via quick-look diagnostics and detailed joint-wise statistics reports

The high-quality data enabled E&P companies to plan remedial work with greater confidence, reducing unnecessary interventions and ensuring more targeted, cost-effective repairs.

Conclusions

The introduction of a high-resolution dual caliper tool with 80 fingers and a compact body design represents a significant advancement in pipe evaluation diagnostics. By enabling single-pass evaluation across a broad range of pipe sizes and offering superior defect resolution, this technology directly addresses limitations associated with traditional multifinger calipers. Its deployment in real-world conditions has proven its robustness, versatility, and value in delivering actionable insights for E&P companies.

This work presents a transformative approach to tubular evaluation by integrating extended-range logging with high-resolution defect detection in a single deployment. The ability to assess multiple casing sizes in one descent not only enhances operational efficiency but also empowers E&P companies to make faster, more informed decisions. The tool's adaptability and precision make it particularly valuable for managing aging infrastructure and complex completions.

References

- Dennis, Bob. "Casing Corrosion Evaluation Using Wireline Techniques." *J Can Pet Technol* **29** (1990). doi: <https://doi.org/10.2118/90-04-09>